THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS UNIT TEST SET B

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. The product of any number and 0 is: 1

(a) 0 **(b)** undefined
(c) the number itself **(d)** 1

2. The common mistake in $(a+b)^2 = a^2 + b^2$ is: 1

(a) wrong sign **(b)** missing the term $2ab$
(c) extra a **(d)** none of these

3. Rohan is 12 years old and his father is 42. The ratio of their ages is: 1

(a) 7 : 2 **(b)** 1 : 3 **(c)** 2 : 7 **(d)** 3 : 7

4. The ratio 18 : 24 in its simplest form is: 1

(a) 3 : 4 **(b)** 4 : 4 **(c)** 4 : 3 **(d)** 3 : 5

5. Two ratios are equal if they: 1

(a) have the same value when simplified **(b)** have the same sum
(c) have the same first term **(d)** have equal differences

6. A gain of 100% means the selling price is: 1

(a) twice the cost price **(b)** equal to cost price
(c) 10% more **(d)** half of cost price

7. An article bought for ₹ 500 is sold for ₹ 575. The profit percent is: 1

(a) 20% **(b)** 15% **(c)** 75% **(d)** 7.5%

8. $\frac{1}{16}$ expressed as a percentage is: 1

(a) 11.25% **(b)** 12.5% **(c)** 1.25% **(d)** 6.25%

9. The legs of a right triangle are 5 cm and 12 cm. Its hypotenuse is: 1

(a) 13 cm **(b)** 14 cm **(c)** 17 cm **(d)** 12 cm

10. Baudhayana's Shulba Sutra describes how to: 1

(a) write numbers
(b) calculate profit
(c) measure time
(d) construct altars using rectangles and squares

11. 9 workers finish a job in 16 days. How many days will 12 workers take? 1

(a) 21.3333 **(b)** 11 **(c)** 14 **(d)** 12

12. On a map, 1 cm represents 50 km. A distance of 2.4 cm on the map represents: 1

(a) 240 km **(b)** 60 km **(c)** 170 km **(d)** 120 km

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 10–15 words each.

13. Is $(a+b)^2 = a^2 + b^2$? Verify with $a = 2$, $b = 3$. 2

14. Are 4 : 9 and 12 : 27 in proportion? Why? 2

15. Find the simple interest on ₹ 6000 at 5% for 3 years. 2

16. State the Baudhayana–Pythagoras theorem. 2

17. A pie chart sector is 72°. What percent of the whole is it? 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 30–40 words each.

18. Evaluate $41^2 - 39^2$ without calculating each square. 3

19. Two numbers are in the ratio 7 : 3 and their sum is 50. Find the numbers. 3

20. A rectangular park measures 3 m by 4 m. A boy walks along the diagonal instead of two sides. How much distance does he save? 3

21. Divide ₹ 4400 among A, B and C in the ratio 1:3:7. 3

22. On a map, 1 cm stands for 40 km. Two cities are 7.5 cm apart. Find the actual distance. How long does a train at 60 km/h take? 3

OR

8 workers can complete a task in 45 days. How many days will 18 workers take, and how many extra days if 8 had worked? 3

23. Find 1002^2 using a suitable identity. 3

OR

Expand and simplify $(x+4)(x+11)$. Verify for $x = 1$. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 80–100 words each.

24. (i) Prove $(a+b)(a-b) = a^2 - b^2$. 5

(ii) Compute 52×48 and 103×97 using it.

OR

(i) Derive $(a+b)^2$ using an area model. 5

(ii) Evaluate 73^2 using it.

25. (i) The population of a village rose from 8000 to 9680 in 2 years at a constant yearly rate. Find the rate. 5

(ii) Predict next year's.

OR

(i) Find the CI on ₹ 20000 at 10% p.a. for 3 years. 5

(ii) Compare with the SI.

END OF QUESTION PAPER Total: 50 Marks

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